Bayesian Multiclass Segmentation for Remote Sensing: Integrating User Priors and Uncertainty

Yeongsu Kim, Haeyun Lee, Seo-Yeon Choi, and Kyungsu Lee

***Abstract*—Geospatial object segmentation from aerial imagery is a significant task in remote sensing, underpinning a wide range of applications such as environmental monitoring, urban planning, and defense. Despite advances in deep learning (DL), conventional segmentation methodologies often suffer from limited generalizability due to regional variability, data distribution shifts, and ambiguous object boundaries. Moreover, most existing approaches are based on static, pretrained architectures with the lack of adaptability to unseen domains or user-specific requirements. To alleviate this, we introduce, in this study, a novel interactive segmentation framework that integrates few-shot semionline adaptation with a Bayesian variational inference mechanism. Our approach dynamically incorporates user corrections and convolutional feature representations to construct an adaptive prior within a variational autoencoder (VAE), thereby enabling user knowledge and feedback to be directly encoded into the segmentation process. This probabilistic formulation enhances the iterative refinement of segmentation masks in response to user input and enables principled quantification of model uncertainty, which is a strong factor for trustworthy deployment in real-world geospatial applications. Extensive experiments on multiple benchmark remote sensing datasets demonstrate that our method achieves superior segmentation accuracy and robustness with minimal user intervention, outperforming existing test-time adaptation (TTA) and interactive segmentation techniques. The proposed framework represents a scalable and user-adaptive solution, bridging the gap between automated segmentation and domain-expert guidance, and advancing the state of the art in geospatial analysis for heterogeneous and dynamic environments.**

***Index Terms*—Aerial image segmentation, Bayesian inference, interactive learning, remote sensing, uncertainty quantification.**

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I. Introduction

GEOSPATIAL object segmentation is a fundamental task in remote sensing, supporting a wide spectrum of applications such as urban planning, disaster response, land-cover monitoring, and environmental management [1], [2], [3], [4]. In recent years, deep-learning (DL)-based methods have substantially advanced the state of the art in multiclass semantic segmentation of aerial imagery by exploiting large-scale annotated datasets. Convolutional neural networks (CNNs) and transformer-based architectures [5], [6] have shown remarkable capacity in modeling spatial relationships and extracting semantic features for a diverse range of land-cover types, including buildings, roads, vegetation, and water bodies [7], [8], [9]. These developments have promoted the automation of geospatial analysis and facilitated large-scale, multiclass segmentation in both urban and rural environments [10], [11], [12], [13].

Nevertheless, robust multiclass segmentation from aerial images remains a challenging problem due to several intrinsic factors [14], [15], [16]. A key difficulty lies in the ambiguity of object boundaries and intraclass variability, which are exacerbated by limited spatial resolution, occlusions, seasonal or environmental changes, and complex landscape patterns [17], [18], [19], [20]. Unlike natural objects with well-defined contours, land-cover classes such as vegetation or impervious surfaces often present indeterminate and subjective edges, leading to segmentation results that are highly dependent on application context and annotator expertise. These challenges demand segmentation methodologies that are adaptable to diverse class definitions, user requirements, and heterogeneous geospatial settings [21], [22], [23]. Another persistent limitation is the sensitivity of current DL-based models to domain shifts, which can be caused by variations in sensor characteristics, acquisition conditions, and regional heterogeneity [4], [23]. As a result, models trained on a specific dataset frequently suffer substantial performance drops when transferred to novel or unseen domains [23], [24], [25].

To address these issues, test-time adaptation (TTA) has emerged as an effective strategy, enabling pretrained models to adapt to new data distributions using unlabeled test samples prior to inference [26], [27], [28], [29], [30], [31], [32], [33]. Techniques such as entropy minimization and self-training have been widely explored in image classification [26], [27]

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and, more recently, in multiclass semantic segmentation [28], [29], [30]. However, most TTA approaches rely on rigid adaptation strategies and lack the capacity for user-driven, class-specific refinement, thus limiting their generalization in complex operational scenarios. Online learning frameworks have also been investigated for incremental model updates and real-time adaptation [31], [32], [33]. Yet, these methods often overlook the integration of human expertise and prior knowledge, which is especially critical in multiclass settings where class boundaries can be ambiguous.

To alleviate these limitations, we present a novel interactive segmentation framework specifically designed for multiclass land-cover mapping in remote sensing imagery. In our approach, all the convolutional components are implemented as Bayesian convolutional neural networks (BCNNs) [34], allowing the model to capture and propagate predictive uncertainty throughout the segmentation pipeline. During inference, user-provided correction masks for ambiguous or misclassified regions are explicitly matched to the corresponding spatial features extracted by each BCNN. These matched features are then merged and used to parameterize the adaptive prior distribution in the latent space of a variational autoencoder (VAE) [35]. This design enables the model to flexibly incorporate both visual context and user domain knowledge, with the prior dynamically conditioned on both the input imagery and real-time user feedback. Consequently, each refinement step directly leverages user guidance, allowing the segmentation model to progressively improve accuracy in challenging or subjective regions. Furthermore, the Bayesian formulation naturally supports principled quantification of predictive uncertainty at the pixel level, enabling reliable identification of regions with high ambiguity or low model confidence. This uncertainty information can be exploited to prioritize further user intervention or inform downstream postprocessing, thereby increasing the reliability and trustworthiness of the system for operational deployment. To summarize, our main contributions are as follows.

1) We present an interactive segmentation framework for multiclass land-cover mapping, where all the convolutional modules are replaced by BCNNs to enable robust, uncertainty-aware adaptation.
2) User corrections are directly matched with BCNN features to parameterize the prior in a VAE, effectively incorporating domain expertise into the segmentation process.
3) Experiments on public datasets show that our method outperforms existing adaptation and interactive segmentation in both accuracy and reliability with minimal user input.

II. Related Works

A. Test-Time Adaptation

TTA aims to enable pretrained models to maintain robustness when exposed to distribution shifts at inference, by adapting to unlabeled test data prior to prediction [26]. Early research on TTA focused largely on image classification, using strategies such as entropy minimization, self-training, and adaptation of batch normalization statistics to improve test-time generalization [27]. More recently, TTA has also shown promise in segmentation tasks. For example, Prabhudesai et al. [28] proposed slot-TTA, leveraging slow inference to boost segmentation performance under out-of-distribution (OOD) conditions. Ma [29] introduced improved self-training (IST), achieving notable gains in both classification and segmentation. Chen et al. [30] further advanced the field by tailoring prompt-based adaptations to individual test images for continual TTA in segmentation. Collectively, these works demonstrate that TTA can substantially enhance model resilience to domain shifts and novel environments, but often lack mechanisms for user-guided or uncertainty-aware refinement.

B. Online Learning

Traditional supervised learning methods typically operate in an offline or batch mode, where models are trained on a static dataset and deployed for inference without further updates. This paradigm incurs high retraining costs when new data emerge, limiting scalability and real-time applicability. In contrast, online learning enables models to incrementally update parameters as new data become available, supporting efficient adaptation in dynamic or continuously changing environments [31]. Online learning has been investigated across a range of domains, including segmentation. For instance, Zhao et al. [32] developed an online-learning-based approach for hand segmentation, while Volpi et al. [33] proposed a protocol for frame-by-frame continual semantic segmentation from image sequences. These studies highlight the potential of online learning for real-time adaptation and performance improvement in nonstationary contexts. However, most prior works do not integrate user feedback or quantify prediction uncertainty, which are important for robust segmentation.

C. Query Fusion

Query-fusion mechanisms constitute a core component of transformer-based segmentation frameworks, in which segmentation is formulated as a set prediction problem driven by a finite collection of query embeddings. Let $Q^{(l)} = \{\mathbf{q}_i^{(l)}\}_{i=1}^{N_q}$ denote the set of queries at decoder layer l, and let $F = \{\mathbf{f}_j\}_{j=1}^{N_f}$ denote the encoded image features. In query-based decoders, each query is refined through attention-based aggregation of spatial and semantic context from the feature space. This process is typically expressed using cross-attention as

$$\tilde{\mathbf{q}}_i^{(l)} = \sum_{j=1}^{N_f} \alpha_{ij}^{(l)} \mathbf{W}_v \mathbf{f}_j \tag{1}$$

where the attention weights are computed as

$$\alpha_{ij}^{(l)} = \text{softmax}\left(\frac{\left(\mathbf{W}_q \mathbf{q}_i^{(l)}\right)^\top \left(\mathbf{W}_k \mathbf{f}_j\right)}{\sqrt{d}} \right) \tag{2}$$

with learnable projection matrices $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$, and d denoting the query embedding dimension. The query is then updated via a residual connection

$$\mathbf{q}_i^{(l+1)} = \mathbf{q}_i^{(l)} + \tilde{\mathbf{q}}_i^{(l)} \tag{3}$$

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allowing query embeddings to progressively integrate global contextual information distributed across the spatial feature map. Within this formulation, query fusion refers to operations that aggregate multiple query representations, either originating from different sources or produced at different decoding stages, into a unified query set that is jointly refined in subsequent layers. Beyond feature-to-query attention, recent studies have demonstrated that fusing queries across decoder layers can mitigate information loss and improve recall in deep decoders. In hierarchical query-fusion strategies, predicted masks from consecutive layers are compared to identify complementary queries. Given mask predictions $\{M_i^{(l)}\}$ at layer l and $\{M_j^{(l-1)}\}$ at layer $l-1$, mask-level overlap scores such as

$$\text{IoU}\left(M_i^{(l)}, M_j^{(l-1)}\right) \tag{4}$$

are used to select informative queries from the previous layer. Queries whose overlap falls below a predefined threshold τ are retained and merged with the current query set as

$$\hat{Q}^{(l)} = Q^{(l)} \cup \left\{ \mathbf{q}_j^{(l-1)} \;\middle|\; \text{IoU}\left(M_i^{(l)}, M_j^{(l-1)}\right) < \tau \right\} \tag{5}$$

and the fused query set $\hat{Q}^{(l)}$ is propagated to deeper decoder layers for joint refinement [36]. By explicitly aggregating query representations through attention and fusion operations, transformer-based segmentation models can effectively capture long-range dependencies and preserve complementary information across decoding stages. As a result, query-fusion mechanisms have become a fundamental design principle in both semantic and instance segmentation, offering a structured alternative to conventional feature-only aggregation strategies.

D. Interactive Image Segmentation

Interactive image segmentation aims to enhance segmentation outcomes by incorporating user-provided cues or prior knowledge to distinguish target regions from the background [37]. Liu et al. [38] proposed an interactive segmentation model based on Vision Transformers, using user clicks as guidance. Maninis et al. [39] introduced the use of extreme points to facilitate segmentation in both the static images and videos, while Zhang et al. [40] developed the inside–outside guidance framework to obtain precise object masks with minimal human interaction [41], [42], [43]. Although these methods improve accuracy through user interaction, they generally focus on single-image refinement and lack mechanisms for accumulating user feedback across multiple predictions or sessions. In contrast, our work introduces a framework that leverages user-modified predictions as auxiliary information for subsequent inference, thereby enabling progressive and cumulative performance improvement without the need for retraining. Unlike prior interactive segmentation approaches that treat user inputs as deterministic corrections or guidance signals applied at the output level, our framework models user interactions as probabilistic priors that condition the latent distribution of the segmentation model. This distinction enables user inputs to influence predictions in a soft and uncertainty-aware manner, rather than enforcing hard constraints.

III. Methods

A. Problem Definition

Let $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ denote a remote sensing image with spatial resolution $H \times W$ and C spectral channels, where typically $C = 3$ for RGB aerial imagery. The primary objective is to predict a multiclass semantic segmentation mask $\mathbf{Y} \in \{1,\ldots,K\}^{H\times W}$, assigning each pixel (i, j) to one of K land-cover categories relevant to geospatial analysis. In this framework, user-provided interaction exploits correction masks, denoted by $\mathcal{U}$, which highlight regions of interest or amend misclassified areas in the segmentation output. The goal is to construct a segmentation model that adaptively integrates the visual information from $\mathbf{I}$ with user bias encoded in $\mathcal{U}$, and also produces calibrated uncertainty estimates for each prediction. Formally, the objective of this study is to model the posterior distribution

$$p(\mathbf{Y} \mid \mathbf{I}, \mathcal{U}) \tag{6}$$

where the output segmentation mask by the model is designed to be both accurate and uncertainty-aware, reflecting the correction bias of the user. In this formulation, the user correction U is treated as an observed variable that conditions the latent prior of the segmentation model, rather than as a deterministic constraint on the output. The objective is therefore to infer a posterior predictive distribution that integrates both the visual evidence and user-provided prior knowledge in a probabilistic manner.

B. Overall Architecture

As illustrated in Fig. 1, the proposed framework is composed of two main stages: a training stage and an inference stage with user interaction. This dual-stage design allows the model to first learn robust feature representations from annotated images, and then adaptively refine segmentation predictions by incorporating user feedback during inference.

In both the stages, the input image is processed by a BCNN encoder, which extracts spatial feature maps that capture the visual context. During inference, if the initial segmentation contains ambiguous or misclassified regions, the user can provide correction masks to indicate these areas. The user-provided corrections are aligned with the BCNN feature maps and integrated into the model pipeline as additional guidance for further refinement.

The extracted features and user corrections are projected into token representations and processed by a transformer encoder, which fuses global context with user-driven information. This enables the model to propagate the effects of user feedback throughout the feature space. The fused token representations are then fed into a VAE module, where user information is incorporated into the latent space prior. This design allows the model to iteratively improve segmentation outputs in response to user input, while also providing pixel-wise uncertainty estimates.

Overall, the architecture supports both the automated and interactive segmentation refinement, enabling robust, uncertainty-aware, and user-adaptive performance. The following sections describe the key components of the framework

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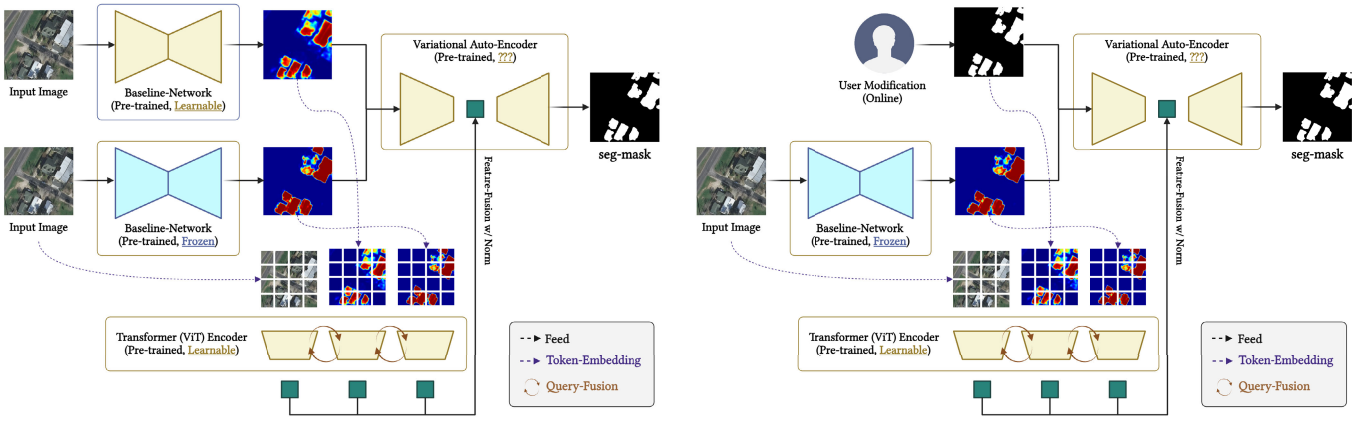


Fig. 1. Overview of the proposed user-adaptive Bayesian segmentation framework. A baseline Bayesian CNN extracts uncertainty-aware features, which are tokenized and refined by a transformer encoder with query fusion to aggregate global context. The refined queries are processed by a VAE for probabilistic segmentation refinement. The annotation "pretrained, ???" indicates that the corresponding modules are trained offline and may operate in different modes depending on the stage. During training, these modules are learnable and optimized end-to-end. During inference and user interaction, they can either be fine-tuned or kept fixed; in the standard setting adopted in this work, all the backbone components are kept fixed, and user corrections are incorporated solely by conditioning the latent prior of the VAE, without updating network parameters.

in detail: the Bayesian CNN backbone, the transformer-based query fusion, and the VAE-driven user-guided adaptation.

C. Bayesian CNN: Epistemic Uncertainty and User Prior

The VAE encoder in our framework is instantiated as a BCNN, in which each Bayesian convolutional layer's weights $\mathbf{W}_l$ are modeled as random variables rather than fixed parameters [34]. During training, an approximate posterior $q(\mathbf{W}_l|\mathcal{D})$ is learned for each layer l given the dataset $\mathcal{D}$. The feature extraction process for input $\mathbf{I}$ is thus probabilistic

$$\mathbf{F} = f_{\text{BCNN}}\left(\mathbf{I};\ \{\mathbf{W}_l\}\right), \quad \mathbf{W}_l \sim q\left(\mathbf{W}_l|\mathcal{D}\right) \quad \forall l. \tag{7}$$

The predictive distribution of the feature map is obtained by marginalizing over the weight posteriors

$$p\left(\mathbf{F}|\mathbf{I}\right) = \int p\left(\mathbf{F}|\mathbf{I}, \{\mathbf{W}_l\}\right) \prod_l q\left(\mathbf{W}_l|\mathcal{D}\right)\, d\mathbf{W}_l. \tag{8}$$

The predictive distribution $p(\mathbf{F}|\mathbf{I})$ is efficiently approximated by Monte Carlo sampling [44]: M feature realizations $\{\mathbf{F}^{(m)}\}_{m=1}^{M}$ are obtained by sampling weights $\mathbf{W}_l \sim q(\mathbf{W}_l|\mathcal{D})$, and the pixelwise empirical variance $\text{Var}_m[\mathbf{F}^{(m)}]$ quantifies epistemic uncertainty. This uncertainty regularizes the segmentation and identifies ambiguous regions that benefit from user intervention.

Importantly, the BCNN encoder in our architecture explicitly incorporates spatial prior information during inference. Particularly, for each user correction mask $\mathbf{M}_{\text{user}}^{(n)} \in \{0,1\}^{H\times W} (n = 1, \ldots, N)$, the BCNN receives both the image feature map $\mathbf{F}$ and a corresponding spatial prior $\mathbf{F}_{\text{prior}}^{(n)}$ as input, where

$$\mathbf{F}_{\text{prior}}^{(n)} = \mathbf{F} \odot \mathbf{M}_{\text{user}}^{(n)}. \tag{9}$$

In the forward pass of the Bayesian convolutional layer, this prior is concatenated or combined with the data-driven feature map, yielding

$$\mathbf{F}_{\text{BCNN}}^{(n)} = \text{BayesConv}\left(\mathbf{F},\ \mathbf{F}_{\text{prior}}^{(n)}; \mathbf{W}\right) \tag{10}$$

where BayesConv$(\cdot)$ denotes the Bayesian convolution operation parameterized by weight samples $\mathbf{W} \sim q(\mathbf{W}|\mathcal{D})$. The resulting user-guided feature maps $\{\mathbf{F}_{\text{BCNN}}^{(n)}\}_{n=1}^{N}$ are then aggregated, such as by spatial pooling or concatenation, and injected as explicit priors into the VAE latent pathway. This mechanism enables multiple user-supplied corrections to directly modulate the generative prior in Eq. 10, thereby allowing domain bias from user feedback to spatially guide and regularize downstream segmentation predictions.

In our Bayesian segmentation framework, the joint probability over all the variables, including the segmentation mask $\mathbf{Y}$, latent variable z, input image $\mathbf{I}$, and a set of user correction masks $\{\mathbf{M}_{\text{user}}^{(n)}\}$, is factorized as

$$\begin{aligned} p\left(\mathbf{Y}, z \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right) &= p\left(\mathbf{Y} \mid z, \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right) \\ &\times\ p\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right) \end{aligned} \tag{11}$$

where $p(\mathbf{Y} \mid z, \mathbf{I}, \{\mathbf{M}_{\text{user}}^{(n)}\})$ denotes the segmentation likelihood and $p(z \mid \mathbf{I}, \{\mathbf{M}_{\text{user}}^{(n)}\})$ is a user-guided prior with aggregated features $\{\mathbf{F}_{\text{prior}}^{(n)}\}$. Since exact inference of $p(z \mid \mathbf{I}, \{\mathbf{M}_{\text{user}}^{(n)}\}, \mathbf{Y})$ is intractable, we use a variational approximation q_ϕ and maximize the evidence lower bound (ELBO)

$$\begin{aligned} &\log p\left(\mathbf{Y} \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right) \\ &\geq -\text{KL}\left(q_\phi\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}, \mathbf{Y}\right) \,\|\, p\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right)\right) \\ &\quad + \mathbb{E}_{q_\phi\left(z|\mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}, \mathbf{Y}\right)}\left[\log p_\theta\left(\mathbf{Y} \mid z, \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right)\right] \end{aligned} \tag{12}$$

where θ and ϕ denote the generative and variational parameters, respectively, and KL indicates KL-divergence. Here, the user correction masks act as auxiliary observed variables that inject spatial evidence into the latent prior. This joint probabilistic framework regularizes segmentation predictions by integrating both epistemic uncertainty, captured by the BCNN encoder, and domain expertise from user corrections, thereby enabling robust and uncertainty-aware segmentation even in challenging or underrepresented scenarios.

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D. Token Embedding and Transformer-Based Query Fusion

After BCNN feature extraction and user-guided prior construction, both the image-driven features $\mathbf{F}$ and user-prior features $\{\mathbf{F}_{\text{prior}}^{(n)}\}$ are projected to token embeddings

$$\mathbf{T}_F = h_F(\mathbf{F}), \quad \mathbf{T}_{\text{prior}}^{(n)} = h_{\text{prior_emb}}\left(\mathbf{F}_{\text{prior}}^{(n)}\right) \tag{13}$$

where h_F and $h_{\text{prior_emb}}$ are learnable linear mappings. These tokens are concatenated to form the input query sequence as

$$\mathbf{Q}^0 = \text{concat}\left(\mathbf{T}_F,\ \left\{\mathbf{T}_{\text{prior}}^{(n)}\right\}_{n=1}^{N}\right) \in \mathbb{R}^{M\times d} \tag{14}$$

where $M = N_{\text{img}} + N$ (image and user tokens), and d is the embedding dimension. A stack of L transformer layers then iteratively refines the queries. At each layer l and for each query i, the update as $\mathbf{q}_i^{(l+1)} = \mathbf{q}_i^{(l)} + \text{MHAttn}(\mathbf{q}_i^{(l)}, \mathbf{K}^{(l)}, \mathbf{V}^{(l)})$, where multihead self-attention is given by

$$\text{MHAttn}(\mathbf{q},\mathbf{K},\mathbf{V}) = \sum_{h=1}^{H} \text{softmax}\left(\frac{\mathbf{q}\,(\mathbf{K}_h)^\top}{\sqrt{d_k}}\right)\mathbf{V}_h \tag{15}$$

with H heads, and $\mathbf{K}_h, \mathbf{V}_h$ being head-specific linear projections of all the queries. The output after L layers

$$\mathbf{Q}^* = \left\{\mathbf{q}_i^{(L)}\right\}_{i=1}^{M} \tag{16}$$

encodes a globally contextual, uncertainty- and user-bias aware representation for downstream decoding. The transformer-based query fusion can be interpreted as a probabilistic message passing scheme over the spatial graph of the image. At each transformer layer l, the update of query i is given by

$$\mathbf{q}_i^{(l+1)} = \mathbf{q}_i^{(l)} + \sum_{j=1}^{M} a_{ij}^{(l)}\,\mathbf{W}_V\mathbf{q}_j^{(l)} \tag{17}$$

where the attention weights $a_{ij}^{(l)}$ are defined as

$$a_{ij}^{(l)} = \frac{\exp\left(\left(\mathbf{q}_i^{(l)}\mathbf{W}_Q\right)\left(\mathbf{q}_j^{(l)}\mathbf{W}_K\right)^\top / \sqrt{d_k}\right)}{\sum_{j'}\exp\left(\left(\mathbf{q}_i^{(l)}\mathbf{W}_Q\right)\left(\mathbf{q}_{j'}^{(l)}\mathbf{W}_K\right)^\top / \sqrt{d_k}\right)}. \tag{18}$$

Here, each $a_{ij}^{(l)}$ encodes the adaptive connectivity (edge weight) between node i and node j in the spatial query graph, conditioned on both the local features and user-provided corrections present in the query embeddings. Recursively applying this update over L layers yields

$$\mathbf{q}_i^* = \mathbf{q}_i^{(L)} = \mathbf{q}_i^{(0)} + \sum_{l=0}^{L-1}\sum_{j=1}^{M}\left(\prod_{k=l+1}^{L-1} a_{ij}^{(k)}\right)\mathbf{W}_V\mathbf{q}_j^{(l)}. \tag{19}$$

Therefore, the final query embedding $\mathbf{q}_i^*$ at each location i is a contextually weighted sum of both the image-driven and user-guided tokens, propagated over the attention graph. This mechanism ensures globally consistent, user-adaptive refinement, and is mathematically analogous to belief propagation, where attention weights a_{ij} serve as adaptive edge potentials.

This formulation establishes a direct connection between transformer-based query fusion and prior query-centric segmentation frameworks. In contrast to conventional query-based decoders that rely on cross-attention between learnable queries and pixel-level feature embeddings, the proposed formulation operates entirely in the query space after tokenization. All the image-derived and user-derived information is embedded into a unified query set and refined through self-attention, resulting in a symmetric interaction among heterogeneous query sources. From the perspective of existing query-fusion strategies, the proposed approach corresponds to a source-level query fusion mechanism. Instead of fusing queries across decoder layers or selectively merging queries based on mask-level similarity, all the query sources are preserved and jointly refined from the initial layer. This design choice avoids heuristic query selection or pruning and ensures that user-guided information remains available throughout the entire refinement process. The influence of each query is adaptively regulated by attention weights and subsequently modulated by uncertainty-aware Bayesian inference in the downstream decoder. As a result, the final query representation Q^* encodes not only global contextual relationships among image regions but also structured user preferences propagated over the attention graph. This interpretation aligns the proposed query-fusion module with existing transformer-based segmentation models, while extending them to support probabilistic, user-adaptive refinement without imposing hard constraints or layer-specific query manipulation.

E. VAE With User-Conditioned Prior

Following transformer-based query fusion, we use a conditional VAE to generate the final segmentation mask. In this framework, the latent variable z is regularized by a prior distribution that adaptively incorporates both the image features and user-guided spatial priors. Formally, the prior for z is parameterized as a multivariate Gaussian

$$p\left(z \mid \mathbf{I}, \left\{\mathbf{F}_{\text{prior}}^{(n)}\right\}\right) = \mathcal{N}\left(z \mid \mu_{\text{prior}}, \Sigma_{\text{prior}}\right). \tag{20}$$

Here, h_{prior} is a learnable function (e.g., an MLP), and $\text{GP}(\cdot)$ denotes global pooling. Here, the user prior does not directly alter the segmentation likelihood. Instead, it modulates the prior distribution of the latent variable z by influencing the parameters $(\mu_{\text{prior}}, \Sigma_{\text{prior}})$. As a result, user inputs affect the posterior predictive distribution only through Bayesian inference, allowing uncertain or inconsistent priors to be naturally reflected as increased posterior uncertainty rather than enforced corrections.

During training, the VAE encoder $q_\phi(z \mid \mathbf{I}, \{\mathbf{M}_{\text{user}}^{(n)}\}, \mathbf{Y})$ infers the variational posterior, while the decoder $p_\theta(\mathbf{Y} \mid z, \mathbf{Q}^*)$ reconstructs the segmentation mask. The ELBO objective is given by

$$\begin{aligned}&\log p\left(\mathbf{Y} \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}\right)\\ &\geq \mathbb{E}_{q_\phi\left(z\mid\mathbf{I},\left\{\mathbf{M}_{\text{user}}^{(n)}\right\},\mathbf{Y}\right)}\left[\log p_\theta(\mathbf{Y} \mid z, \mathbf{Q}^*)\right]\\ &\quad - \text{KL}\left(q_\phi\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\text{user}}^{(n)}\right\}, \mathbf{Y}\right) \,\|\, p\left(z \mid \mathbf{I}, \left\{\mathbf{F}_{\text{prior}}^{(n)}\right\}\right)\right).\end{aligned} \tag{21}$$

Let the pooled user-guided features $\mathbf{s}_{\text{user}} = \{\text{GP}(\mathbf{F}_{\text{prior}}^{(n)})\}_{n=1}^{N}$ be denoted as a sufficient statistic for the user correction masks. Under the generative model, the latent prior is as

$$p(z \mid \mathbf{I}, \mathbf{s}_{\text{user}}) = \mathcal{N}\left(\mu_{\text{prior}}, \Sigma_{\text{prior}}\right) \tag{22}$$

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TABLE I
FUNCTIONAL PIPELINES OF THE PROPOSED SEGMENTATION FRAMEWORK FOR TRAINING AND INFERENCE

Order	Input(s)	Output(s)	Description
1. Feature Extraction	$\mathbf{I}$	$\mathbf{F} = f_{\mathrm{BCNN}}(\mathbf{I};\, \mathbf{W})$	Extract deep features with BCNN encoder
2. User Prior Modulation	$\mathbf{F},\ \{\mathbf{M}_{\mathrm{user}}^{(n)}\}$	$\mathbf{F}_{\mathrm{prior}}^{(n)} = \mathbf{F} \odot \mathbf{M}_{\mathrm{user}}^{(n)}$	Inject spatial user prior into feature map
3. Token Embedding	$\mathbf{F},\ \{\mathbf{F}_{\mathrm{prior}}^{(n)}\}$	$\mathbf{Q}^0 = \mathrm{concat}\big(h_F(\mathbf{F}),\ \{h_{\mathrm{prior_emb}}(\mathbf{F}_{\mathrm{prior}}^{(n)})\}\big)$	Project features and priors into query tokens
4. Query Fusion	$\mathbf{Q}^0$	$\mathbf{Q}^* = \mathrm{Transformer}(\mathbf{Q}^0)$	Global context integration via transformer
5. Prior Parameterization	$\mathbf{F},\ \{\mathbf{F}_{\mathrm{prior}}^{(n)}\}$	$[\mu_{\mathrm{prior}},\, \Sigma_{\mathrm{prior}}] = h_{\mathrm{prior}}\big(\mathrm{GP}(\mathbf{F}),\ \{\mathrm{GP}(\mathbf{F}_{\mathrm{prior}}^{(n)})\}\big)$	Compute VAE prior with user guidance
6. Latent Sampling	$[\mu_{\mathrm{prior}},\, \Sigma_{\mathrm{prior}}]$	$z \sim \mathcal{N}(\mu_{\mathrm{prior}},\, \Sigma_{\mathrm{prior}})$	Sample latent variable for generative pathway
7. Decoding	$\mathbf{Q}^*,\ z$	$\mathbf{Y}_{\mathrm{pred}} = f_{\mathrm{dec}}(\mathbf{Q}^*,\, z)$	Produce final segmentation mask

where $[\mu_{\mathrm{prior}}, \Sigma_{\mathrm{prior}}] = h_{\mathrm{prior}}(\mathrm{GP}(\mathbf{F}), \mathbf{s}_{\mathrm{user}})$. The posterior predictive for mask then marginalizes over the latent variable

$$p\left(\mathbf{Y} \mid \mathbf{I},\, \mathbf{s}_{\mathrm{user}}\right) = \int p_\theta\left(\mathbf{Y} \mid z,\, \mathbf{Q}^*\right)\, p\left(z \mid \mathbf{I},\, \mathbf{s}_{\mathrm{user}}\right)\, dz. \tag{23}$$

It follows that by the property of conditional independence in hierarchical Bayesian models, the distribution $p(\mathbf{Y} \mid \mathbf{I}, \mathbf{s}_{\mathrm{user}})$ is adaptively regularized by $\mathbf{s}_{\mathrm{user}}$, a sufficient statistic for domain knowledge injected via user correction, and any change in $\mathbf{s}_{\mathrm{user}}$ directly modulates the prior and, therefore, the entire predictive distribution $p(\mathbf{Y} \mid \cdot)$. Thus, segmentation outputs are guaranteed to be user-adaptive and uncertainty-aware, as both the observed data and the expert-provided spatial evidence regularize the joint distribution.

F. Training Pipeline

The proposed novel framework is organized as a sequence of functional modules. Table I summarizes the full pipeline for both training and inference. During training, ground-truth masks $\mathbf{Y}$ are available, and the model is optimized with a composite objective comprising a supervised segmentation loss and a variational regularization term

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{sup}} + \lambda_{\mathrm{ELBO}}\, \mathcal{L}_{\mathrm{ELBO}} \tag{24}$$

where λ_{ELBO} balances the two terms. The supervised loss $\mathcal{L}_{\mathrm{sup}}$ is defined as

$$\mathcal{L}_{\mathrm{sup}} = \mathbb{E}_{(\mathbf{I},\mathbf{Y})\sim\mathcal{D}}\left[\mathcal{L}_{\mathrm{seg}}\left(\mathbf{Y},\, \mathbb{E}_{q_\phi(z)}\left[p_\theta\left(\mathbf{Y} \mid z, \mathbf{Q}^*\right)\right]\right)\right] \tag{25}$$

where $\mathcal{L}_{\mathrm{seg}}$ is the pixelwise cross-entropy or Dice loss. The variational term is given by the ELBO

$$\begin{aligned}\mathcal{L}_{\mathrm{ELBO}} = {} & \mathbb{E}_{q_\phi\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\mathrm{user}}^{(n)}\right\}, \mathbf{Y}\right)}\left[\log p_\theta\left(\mathbf{Y} \mid z, \mathbf{Q}^*\right)\right] \\ & - \mathrm{KL}\left(q_\phi\left(z \mid \mathbf{I}, \left\{\mathbf{M}_{\mathrm{user}}^{(n)}\right\}, \mathbf{Y}\right) \,\|\, p\left(z \mid \mathbf{I}, \left\{\mathbf{F}_{\mathrm{prior}}^{(n)}\right\}\right)\right).\end{aligned} \tag{26}$$

G. Inference Pipeline

At test time, the inference procedure accommodates two distinct scenarios, depending on whether user-provided correction masks are available for new or unseen datasets: 1) the user provides correction masks in advance, allowing these priors to directly guide segmentation on new or unseen datasets and 2) no initial user corrections are given; the model predicts from the image alone, but if the result is unsatisfactory, the user may supply corrections after inspecting the output.

In the first scenario, suppose a set of N user correction masks $\mathcal{U} = \{\mathbf{M}_{\mathrm{user}}^{(n)}\}_{n=1}^{N}$ is provided in advance. These masks are leveraged as spatial priors to encode user knowledge and domain bias. For each input image $\mathbf{I}$, the user corrections are fused into the latent prior, and M posterior samples $z^{(m)} \sim p(z \mid \mathbf{I}, \mathcal{U})$ are drawn. The corresponding segmentation outputs are computed as

$$\mathbf{S}^{(m)} = f_{\mathrm{dec}}\left(\mathbf{Q}^*,\, z^{(m)}\right), \qquad m = 1, \ldots, M \tag{27}$$

where $\mathbf{Q}^*$ denotes the transformer-fused queries derived from both the image features and user priors in $\mathcal{U}$. The final predicted segmentation mask ($\overline{\mathbf{S}}$) is given by the Monte Carlo mean with the pixelwise epistemic uncertainty ($\mathbf{U}$) as below

$$\begin{aligned}\overline{\mathbf{S}} &= \frac{1}{M}\sum_{m=1}^{M}\mathbf{S}^{(m)} \\ \mathbf{U}(i,j) &= \frac{1}{M-1}\sum_{m=1}^{M}\left(\mathbf{S}^{(m)}(i,j) - \overline{\mathbf{S}}(i,j)\right)^2\end{aligned} \tag{28}$$

where $\mathbf{U}(i, j)$ indicates the confidence for each pixel, with high values signaling ambiguous or error-prone regions, leading to notification for user review or further refinement.

In the second scenario, no prior user corrections are provided initially, i.e., $\mathcal{U} = \emptyset$. The model then operates in a standard automated strategy, where the input image $\mathbf{I}$ is processed without user-conditioned priors. In this scenario, the predictions are generated by sampling $z^{(m)} \sim p(z \mid \mathbf{I})$ as

$$\mathbf{S}^{(m)} = f_{\mathrm{dec}}\left(\mathbf{Q}_{\mathrm{vanilla}}^*,\, z^{(m)}\right), \quad m = 1, \ldots, M \tag{29}$$

where z^{vanilla} is the transformer architecture without the encoder for user preference and $\mathbf{Q}_{\mathrm{vanilla}}^*$ is obtained from image features alone. The mean and variance are computed [see (28)], yielding an uncertainty map $\mathbf{U}(i, j)$ even in the absence of user correction masks. If the resulting segmentation output is unsatisfactory or diverges from user preference, the user may subsequently provide correction masks $\mathcal{U}' = \{\mathbf{M}_{\mathrm{user}}^{(n)}\}_{n=1}^{N'}$, which are then incorporated as explicit spatial priors in the next inference round.

For both the scenarios, the framework supports iterative TTA, where user corrections and model predictions can be alternately updated over $t = 0, 1, \ldots, T_{\max} - 1$

$$\mathbf{Y}_{\mathrm{pred}}^{(t+1)} = f_{\mathrm{dec}}\left(\mathbf{Q}^{*,(t)},\, z^{(t)}\right), \quad z^{(t)} \sim p\left(z \mid \mathbf{I},\, \mathcal{U}^{(t)}\right) \tag{30}$$

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with $\mathcal{U}^{(t)}$ denoting the user correction set at iteration t. The adaptation loop terminates either when

$$\left\| \mathbf{Y}_{\text{pred}}^{(t+1)} - \mathbf{Y}_{\text{pred}}^{(t)} \right\|_1 < \varepsilon \tag{31}$$

for some tolerance $\varepsilon > 0$, or after $t = T_{\max}$ steps.

In practice, the iterative TTA procedure is designed as a practical inference heuristic rather than a theoretically convergent algorithm. Empirically, we observe rapid stabilization of segmentation performance within a small number of refinement steps (typically $T_{\max} = 3$), without oscillatory behavior. We therefore do not claim formal convergence guarantees, but validate the effectiveness of the procedure through empirical performance saturation and risk reduction.

Theoretically, our framework yields a consistent Bayesian estimator of the posterior predictive distribution as below

$$p(\mathbf{Y} \mid \mathbf{I}, \mathcal{U}) = \int p(\mathbf{Y} \mid z, \mathbf{I}, \mathcal{U})\ p(z \mid \mathbf{I}, \mathcal{U})\ dz \tag{32}$$

and the resulting segmentation is Bayes-optimal in the sense of minimizing the expected risk

$$\mathbf{Y}_{\text{pred}} = \arg\min_{\hat{\mathbf{Y}}} \mathbb{E}_{p(\mathbf{Y} \mid \mathbf{I}, \mathcal{U})} \left[\ell\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) \right] \tag{33}$$

where $\ell(\cdot, \cdot)$ is the segmentation loss. The uncertainty map $\mathbf{U}(i, j)$ is consistently defined and computed for both the scenarios, providing a principled means for highlighting ambiguous regions, guiding user interaction, and enabling robust user-adaptive segmentation across operational contexts. We emphasize that the term Bayesian optimal is used here in the classical decision-theoretic sense: the predicted segmentation minimizes the expected loss with respect to the posterior predictive distribution under the assumed probabilistic model. This notion of optimality is therefore conditional on the model family and the variational approximation used and does not imply global convergence guarantees or asymptotic optimality to the true data-generating process.

IV. Experiments

A. Experimental Environment

All the models were implemented in PyTorch (ver. 2.4.0) with CUDA acceleration [45]. Input images were clipped and resized into 512 × 512 tiles, and the mini-batch size was fixed to 8 unless otherwise specified. Optimization was carried out using the AdamW optimizer [46] with an initial learning rate of 1×10^{-4} and polynomial decay. BCNN inference used batch normalization [47] and Monte Carlo dropout sampling with $M = 20$ passes unless explicitly ablated. For interactive experiments, we set the maximum iteration budget to $T_{\max} = 3$. All the experiments were conducted on a server equipped with eight NVIDIA RTX A6000 GPUs (48 GB each).

B. Datasets

To comprehensively evaluate the proposed framework, we considered six benchmark datasets covering diverse remote sensing domains. The DeepGlobe dataset [48] was introduced as part of the CVPR Earth Vision Challenge and provides diverse land-cover categories for large-scale semantic segmentation. The Inria Aerial Image Labeling dataset [49] contains high-resolution aerial imagery with detailed building annotations from multiple cities. LoveDA [4], [50] focuses on land-cover classification under domain adaptation settings, including both urban and rural scenes. UAVid [51] offers semantic segmentation for UAV-based video frames, emphasizing dynamic and street-level environments. OpenEarthMap [52] provides globally distributed, high-resolution satellite imagery for generalized few-shot semantic segmentation. Finally, the Massachusetts Buildings dataset [53] contains aerial imagery with building footprints widely used for building extraction benchmarks. These datasets span different geographic regions, imaging modalities, and spatial resolutions, which allows us to evaluate both generalization and robustness of the proposed Bayesian segmentation framework to domain shift. For all the datasets, we followed the official train/val/test splits when available; otherwise, we adopted a stratified partitioning. To simulate user interactions, we sampled error-prone regions and injected manual corrections according to a predefined interaction budget ($N = 0, 1, 3, 5, 10$ corrections per image). This unified protocol enables quantitative comparisons among static baselines, TTA methods, human-in-the-loop (HITL) refinement models, and our proposed approach.

C. Compared Models

We benchmarked our method against four categories of models. *Baseline models* include U-Net [54], Attention U-Net [55], and UNetPPL [17], which are widely used backbones for remote sensing segmentation. TTA methods comprise CertainTTA [56], SMN [23], and RS-TTA [57], representing adaptation strategies designed for robustness under domain shift. HITL approaches include SAM2 [41], SAM-RS [42], and FCA-Net [43], which integrate user interactions into the segmentation process. Finally, we compared with recent state-of-the-art (SOTA) segmentation models, including RS3Mamba [58], CUPS [59], and TA-Net [60], which have achieved competitive results in remote sensing imagery segmentation benchmarks. This broad selection of baselines ensures a fair and comprehensive evaluation of our framework across static, adaptive, interactive, and SOTA performance.

D. Evaluation Metrics

Performance was primarily assessed with mean intersection-over-union (mIoU), accuracy, and $F1$ score. For a predicted segmentation mask $\hat{Y}$ and ground-truth mask Y, let TP, FP, and FN denote the number of true positives, false positives, and false negatives at the pixel level. The IoU for class c is

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{34}$$

and the mIoU is the average over all the classes. To evaluate calibration, we report both the expected calibration error (ECE) and negative log likelihood (NLL). ECE is computed by first dividing prediction confidences into M bins $\{B_m\}_{m=1}^{M}$. For each bin B_m, we compute the empirical accuracy $\text{acc}(B_m)$

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TABLE II

ABLATION STUDY ON SIX DATASETS IN TERMS OF MIOU (↑). VARIANTS A0–A6 INCREMENTALLY ADD BCNN, TRANSFORMER QUERY FUSION, VAE PRIORS, AND UNCERTAINTY-DRIVEN SELECTION

Variant	DeepGlob	Inria	LoveDA	UAVid	OpenEarth	Mass.	Avg
A0	58.2±0.6	70.4±0.4	46.1±0.8	55.3±0.7	61.2±0.5	72.0±0.3	60.5±0.6
A1	60.1±0.6	71.2±0.5	48.3±0.8	56.1±0.7	62.3±0.5	73.1±0.4	61.9±0.6
A2	62.0±0.5	72.5±0.4	50.2±0.7	58.4±0.6	63.5±0.5	74.2±0.4	63.5±0.5
A3	63.1±0.5	73.3±0.4	51.3±0.7	59.1±0.6	64.4±0.5	75.0±0.4	64.4±0.5
A4	64.0±0.5	74.0±0.4	52.4±0.7	60.0±0.6	65.2±0.5	76.2±0.4	65.3±0.5
A4-Det	63.6±0.5	73.6±0.4	51.8±0.7	59.4±0.6	64.7±0.5	75.9±0.4	64.8±0.5
A5	66.0±0.4	75.1±0.3	54.3±0.6	62.1±0.5	66.3±0.4	77.1±0.3	66.8±0.4
A6	**68.2**±0.4	**76.3**±0.3	**56.0**±0.6	**64.0**±0.5	**68.1**±0.4	**78.3**±0.3	**68.5**±0.4

TABLE III

ABLATION STUDY IN TERMS OF CALIBRATION (ECE ↓) AND EFFICIENCY (FPS, PARAMETER COUNT, $T_{\max}$). THE FULL MODEL (A6) ACHIEVES THE BEST CALIBRATION WHILE MAINTAINING COMPETITIVE EFFICIENCY

Variant	DeepGlob	Inria	LoveDA	UAVid	OpenEarth	Mass.	Avg	FPS	#Params (M)	$T_{\max}$
A0	0.128	0.094	0.156	0.141	0.121	0.088	0.121	58	26	0
A1	0.115	0.090	0.142	0.130	0.112	0.084	0.112	55	27	0
A2	0.108	0.086	0.136	0.125	0.106	0.081	0.107	52	36	0
A3	0.104	0.084	0.131	0.121	0.103	0.079	0.103	51	34	0
A4	0.100	0.082	0.126	0.118	0.100	0.077	0.101	50	35	0
A4-Det	0.112	0.088	0.138	0.128	0.110	0.083	0.110	52	34	0
A5	0.093	0.079	0.118	0.110	0.095	0.074	0.095	48	44	0
A6	**0.086**	**0.075**	**0.110**	**0.103**	**0.089**	**0.070**	**0.089**	46	44	3

and the average confidence conf(B_m). ECE is then given as a weighted average of absolute differences

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| \mathrm{acc}\left(B_m\right) - \mathrm{conf}\left(B_m\right) \right| \tag{35}$$

where n is the total number of predictions. NLL is defined as the average negative log-probability of the true labels under the predicted distribution

$$\mathrm{NLL} = -\frac{1}{n} \sum_{i=1}^{n} \log p\left(y_i \mid x_i\right). \tag{36}$$

We additionally reported computational efficiency in terms of frames per second (FPS) and the number of model parameters (M). For uncertainty evaluation, we plotted risk–coverage curves and reliability diagrams. The risk–coverage curve measures segmentation error as a function of retained prediction coverage when progressively filtering out the most uncertain pixels, while the reliability diagram visualizes calibration by plotting predicted confidence against empirical accuracy.

E. Ablation Study

1) Architectural Study: To better understand the contribution of each component in our framework, we conducted an ablation study across six datasets. Table II reports mIoU performance, while Table III summarizes calibration (ECE) and efficiency (FPS and parameter counts). Starting from the deterministic baseline (A0), the introduction of Bayesian CNN modeling (A1) consistently improved both segmentation accuracy and calibration by enabling explicit uncertainty estimation. Adding the transformer-based query fusion module (A2) further enhanced mIoU by capturing long-range dependencies and global contextual information. Incorporating the VAE-based prior without user corrections (A3) yielded additional gains by regularizing latent representations, while activating user priors (A4) provided more stable improvements in challenging and ambiguous regions. To explicitly isolate the contribution of uncertainty estimation, we further include a deterministic variant (A4-Det) that removes BCNN-based uncertainty modeling while retaining the transformer, VAE, and user priors. Although this variant achieves competitive mIoU in some cases, it exhibits notably degraded calibration performance compared with its Bayesian counterpart (A4), highlighting that uncertainty modeling contributes beyond raw accuracy by improving reliability and risk awareness. The joint use of BCNN, transformer, and VAE with user priors (A5) results in clear and consistent performance gains across all the datasets, while our full model (A6), which additionally integrates uncertainty-driven selection, achieves the best overall results with an average mIoU of 68.5% and the lowest ECE of 0.089.

Beyond accuracy and calibration, we also analyzed efficiency. While the introduction of additional modules slightly increased parameter counts (from 26M in A0 to 44M in A6) and reduced inference speed (from 58 to 46 FPS), the relative decrease in efficiency is modest compared with the substantial improvements in accuracy and reliability. Moreover, the uncertainty-driven selection requires only a small number of additional iterations ($T_{\max} = 3$), demonstrating that the interactive refinement process can be efficiently integrated into real-time workflows. Overall, the ablation results confirm that each component, including Bayesian modeling, query fusion, VAE priors, and uncertainty-driven selection, contributes positively and complementarily to the final performance.

2) TTA and User Prior: We further analyze the joint effect of TTA and user priors across all the six datasets. Each cell reports the average mIoU and ECE. The results show that both the factors improve performance, with diminishing returns beyond three steps or three priors. Table IV shows that both the TTA and user priors steadily improve segmentation accuracy and calibration. The majority of gains are achieved with three priors and three adaptation steps, after which the results saturate near 68.5 mIoU and 0.086 ECE.

TABLE IV

AVERAGE MIOU (%, ↑) AND ECE (↓) ACROSS SIX DATASETS AS A FUNCTION OF TTA STEPS (COLUMNS) AND USER PRIORS N (ROWS)

N \ TTA	0	1	3	5	10
0	66.2 / 0.098	66.8 / 0.095	67.5 / 0.092	67.9 / 0.091	68.1 / 0.090
1	66.9 / 0.096	67.4 / 0.093	68.0 / 0.091	68.3 / 0.090	68.4 / 0.089
3	67.6 / 0.094	68.1 / 0.092	68.4 / 0.090	68.5 / 0.089	68.5 / 0.088
5	68.0 / 0.093	68.4 / 0.091	68.5 / 0.089	68.5 / 0.088	68.5 / 0.087
10	68.2 / 0.092	68.5 / 0.090	68.5 / 0.088	68.5 / 0.087	68.5 / 0.086

F. Overall Segmentation Performance

To ensure fair comparison, all the models were trained and evaluated under identical settings. For static baselines (U-Net [54], Attention U-Net [55], UNetPPL [17]) and SOTA segmentation architectures (RS3Mamba [58], CUPS [59], LANet [60]), we trained using dataset using the papers' official configurations when available, or reproduced them

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TABLE V

OVERALL SEGMENTATION PERFORMANCE (MIOU ↑, MEAN±STD) ACROSS SIX DATASETS WITH EFFICIENCY. LEFTMOST COLUMN SHOWS VERTICAL GROUP LABELS. BASELINES AND SOTA ARE REPORTED SEPARATELY; TTA ADAPTS WITHOUT LABELS, HITL USES SIMULATED USER CORRECTIONS

Family	Method	DeepGlob	Inria	LoveDA	UAVid	OpenEarth	Mass.	Avg	FPS	#Params (M)
Baseline	U-Net [54]	61.0 ± 0.6	72.0 ± 0.5	48.0 ± 0.7	57.0 ± 0.6	62.0 ± 0.5	73.0 ± 0.4	62.2 ± 0.6	65	25
	Attention U-Net [55]	62.0 ± 0.6	72.5 ± 0.5	49.0 ± 0.7	57.5 ± 0.6	63.0 ± 0.5	73.5 ± 0.4	62.9 ± 0.6	63	28
	UNetPPL [17]	62.5 ± 0.5	73.0 ± 0.5	50.0 ± 0.7	58.0 ± 0.6	63.5 ± 0.5	74.0 ± 0.4	63.5 ± 0.6	61	30
SOTA	RS3Mamba [58]	64.0 ± 0.5	74.0 ± 0.4	51.0 ± 0.6	59.0 ± 0.5	64.5 ± 0.5	75.0 ± 0.4	64.6 ± 0.5	59	32
	CUPS [59]	64.5 ± 0.5	74.3 ± 0.4	51.5 ± 0.6	59.2 ± 0.5	65.0 ± 0.5	75.2 ± 0.4	64.9 ± 0.5	58	33
	LANet [60]	65.0 ± 0.5	74.5 ± 0.4	52.0 ± 0.6	59.5 ± 0.5	65.3 ± 0.5	75.5 ± 0.4	65.3 ± 0.5	57	34
TTA	CertainTTA [56]	65.0 ± 0.5	74.8 ± 0.4	52.5 ± 0.6	60.0 ± 0.5	65.8 ± 0.5	76.0 ± 0.4	65.7 ± 0.5	53	36
	SMN [23]	65.5 ± 0.5	75.0 ± 0.4	52.8 ± 0.6	60.2 ± 0.5	66.0 ± 0.5	76.2 ± 0.4	66.0 ± 0.5	52	36
	RS-TTA [57]	65.8 ± 0.5	75.2 ± 0.4	53.0 ± 0.6	60.5 ± 0.5	66.2 ± 0.5	76.5 ± 0.4	66.2 ± 0.5	51	37
HITL	SAM2 [41]	66.0 ± 0.5	75.5 ± 0.4	54.0 ± 0.6	61.0 ± 0.5	66.8 ± 0.5	77.0 ± 0.4	66.7 ± 0.5	50	40
	SAM-RS [42]	66.2 ± 0.5	75.7 ± 0.4	54.3 ± 0.6	61.2 ± 0.5	67.0 ± 0.5	77.2 ± 0.4	66.9 ± 0.5	49	41
	FCA-Net [43]	66.5 ± 0.5	76.0 ± 0.4	54.5 ± 0.6	61.5 ± 0.5	67.3 ± 0.5	77.5 ± 0.4	67.2 ± 0.5	48	42
Ours	Ours (Full)	68.2 ± 0.4	76.3 ± 0.3	56.0 ± 0.6	64.0 ± 0.5	68.1 ± 0.4	78.3 ± 0.4	68.5 ± 0.4	46	44

following their released code under the same schedule as our framework. TTA methods (CertainTTA [56], SMN [23], RS-TTA [57]) were evaluated using their official codebases with adaptation enabled during inference. To ensure consistency, we allowed each model a fixed adaptation budget of ten gradient steps per image (batch size 1), updating the trainable layers specified in the original implementations; reported results therefore reflect the performance after full TTA rather than zero-shot inference. For HITL approaches (SAM2 [41], SAM-RS [42], FCA-Net [43]), we simulated user interactions by randomly sampling error regions according to a predefined budget $N \in \{0, 1, 3, 5, 10\}$ per image. Candidate correction points were selected by ranking pixels with the highest model uncertainty and injecting synthetic user edits in those regions, consistent with the interaction mode of each method (prompt points/masks for SAM-based models; guidance maps for FCA-Net). Each method was permitted up to three iterative refinements ($T_{\max} = 3$), and the final performance reported in our tables corresponds to the maximum interaction budget. Our full model (A6) integrates BCNN uncertainty, transformer-based query fusion, VAE priors, and uncertainty-driven selection, evaluated under the same computational budget. Across all the methods, throughput (FPS) was measured at batch size 1 on a single A6000 GPU, excluding data loading, and parameter counts reflect the number of trainable weights used at inference.

Table V summarizes the overall segmentation results (mIoU, mean±std) across six benchmark datasets under standard within-domain evaluation. Baseline models (U-Net, Attention U-Net, UNetPPL) serve as reference points and highlight the difficulty of the task, while recent SOTA architectures (RS3Mamba, CUPS, LANet) achieve clear improvements owing to stronger feature representation and spatial modeling. TTA methods (CertainTTA, SMN, RS-TTA) further enhance performance by adapting online without requiring labels, showing the benefit of test-time flexibility. HITL approaches (SAM2, SAM-RS, FCA-Net) provide additional gains by incorporating simulated user corrections, particularly

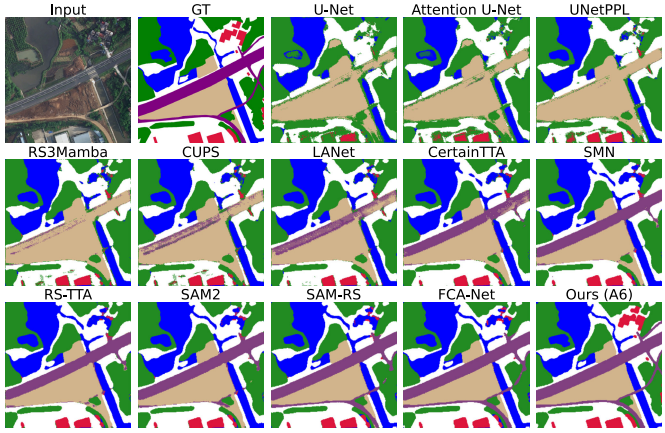


Fig. 2. Representative segmentation results for comparative analysis across different models.

improving boundary-sensitive categories. Our full framework achieves the highest mIoU across all the datasets, surpassing both the static and adaptive baselines, while preserving efficiency in terms of FPS and parameter size, thus demonstrating the effectiveness of integrating Bayesian uncertainty, transformer-based query fusion, and user priors in a unified design. In addition, Fig. 2 represents the segmentation results of all the models and our model exhibits outstanding segmentation performance.

Table VI presents classwise IoU results on the LoveDA dataset, covering seven semantic categories. Baseline models such as U-Net, Attention U-Net, and UNetPPL show limited performance, particularly for Building and Barren classes, which are challenging due to ambiguous boundaries. Recent SOTA models (RS3Mamba, CUPS, LANet) deliver consistent improvements across all the categories, reflecting stronger feature representation. TTA methods further improve results without labels, with the largest gains observed in Water and Forest, highlighting their ability to adapt to distribution shifts. HITL approaches, including SAM2, SAM-RS, and

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TABLE VI

CLASSWISE IOU (%) ON THE LOVEDA DATASET. SEVEN SEMANTIC CATEGORIES ARE REPORTED: BUILDING, ROAD, WATER, BARREN, FOREST, AGRICULTURE, AND BACKGROUND

Method	Building	Road	Water	Barren	Forest	Agriculture	Background	Avg
U-Net [54]	42.1	49.3	52.7	44.8	47.5	48.2	47.3	48.0
Attention U-Net [55]	43.0	50.1	53.5	45.6	48.2	49.0	48.1	49.0
UNetPPL [17]	44.2	51.0	54.3	46.5	49.0	50.0	49.1	50.0
RS3Mamba [58]	45.0	52.2	55.4	47.1	50.2	51.3	49.8	51.0
CUPS [59]	45.5	52.8	55.9	47.7	50.8	51.9	50.4	51.5
LANet [60]	46.0	53.3	56.5	48.2	51.4	52.5	51.0	52.0
CertainTTA [56]	46.7	53.9	57.1	49.0	52.0	53.1	51.6	52.5
SMN [23]	47.0	54.3	57.4	49.3	52.5	53.5	52.1	52.8
RS-TTA [57]	47.3	54.6	57.7	49.6	52.8	53.8	52.5	53.0
SAM2 [41]	48.2	55.2	58.6	50.4	53.5	54.6	53.1	54.0
SAM-RS [42]	48.5	55.5	58.9	50.8	53.8	55.0	53.5	54.3
FCA-Net [43]	49.0	56.0	59.5	51.2	54.3	55.5	54.0	54.5
Ours (Full)	**50.5**	**57.5**	**61.0**	**52.8**	**55.8**	**57.0**	**55.5**	**56.0**

FCA-Net, benefit from simulated user corrections and yield better boundary preservation, especially on Building and Road. Our full framework achieves the best IoU across all the classes, with notable gains on Building, Road, and Agriculture, and reaches an average IoU of 56.0, surpassing all the baselines and aligning with the overall trends in Table V.

G. Domain Adaptation Setting

We follow an unsupervised domain adaptation protocol in which labels are available only for the source domain and the target domain remains unlabeled during adaptation and testing. Four transfers are considered: Inria→Massachusetts, LoveDA Urban→Rural, DeepGlob→OpenEarthMap, and OpenEarthMap→DeepGlob. Source-only baselines and SOTA models are trained on the source split with the same schedule as in the within-domain experiments (input 512, batch size 1, AdamW with initial learning rate 1×10^{-4}) and evaluated directly on the target without adaptation. TTA methods (CertainTTA, SMN, RS-TTA) are run with their publicly released code and objectives; each image receives a fixed budget of ten unlabeled adaptation steps at test time (batch size 1) and early stopping with patience 3 on the unsupervised loss. HITL methods (SAM2, SAM-RS, FCA-Net) are evaluated on the unlabeled target using simulated interactions consistent with the main protocol: an interaction budget N0, 1, 3, 5, 10 per image, uncertainty-ranked edit locations, and at most three refinement rounds; the table reports results at the maximum interaction budget. Our full model (A6) uses the same constraints on the target side, combining Bayesian uncertainty, query fusion, and user priors; unless noted, Monte Carlo sampling uses M = 20 passes. All the numbers are mean±std over three seeds and reported in mIoU.

Table VII shows the expected degradation under domain shift for source-only and SOTA models (average 49.1–52.1), while TTA recovers part of the gap without labels (up to 53.8 on average) and HITL yields additional improvements with limited user input (up to 55.9). The proposed method attains the highest average mIoU of 57.6 across the four transfers, outperforming the best TTA by +3.8 and the best HITL by +1.7 on average. Per-transfer gains over the strongest HITL baseline are consistent: +1.7 on Inria→Massachusetts (62.5 versus 60.8), +1.9 on LoveDA U→R (50.5 versus 48.6), +1.7 on DeepGlob→OpenEarth (59.5 versus 57.8), and +1.7 on OpenEarth→DeepGlob (58.0 versus 56.3). These results indicate that integrating uncertainty with user priors and query fusion is effective for mitigating domain shift while respecting the unlabeled-target constraint.

TABLE VII

DOMAIN ADAPTATION RESULTS (MIOU ↑) ON FOUR SOURCE →TARGET

Method	Inria ↓ Mass.	LoveDA U→R	DeepGlob ↓ OpenEarth	OpenEarth ↓ DeepGlob	Avg
U-Net [54]	54.0	41.5	51.0	49.8	49.1
Attention U-Net [55]	54.8	42.3	51.8	50.5	49.9
UNetPPL [17]	55.6	43.1	52.6	51.2	50.6
RS3Mamba [58]	56.3	43.6	53.2	52.0	51.3
CUPS [59]	56.8	44.0	53.6	52.4	51.7
LANet [60]	57.2	44.5	54.0	52.8	52.1
CertainTTA [56]	58.0	45.8	55.0	53.8	53.2
SMN [23]	58.4	46.1	55.3	54.1	53.5
RS-TTA [57]	58.7	46.5	55.6	54.4	53.8
SAM2 [41]	60.1	47.9	57.1	55.6	55.2
SAM-RS [42]	60.5	48.3	57.5	56.0	55.6
FCA-Net [43] height	60.8	48.6	57.8	56.3	55.9
Ours (Full, A6)	62.5	50.5	59.5	58.0	57.6

H. Similarity of User Priors

We quantify how consistent human priors are across annotators when the same image receives N user hints. For each image, three independent annotators provide $N \in \{1, 3, 5, 10\}$ priors in the same interaction mode used by the methods (points, small polygons, or scribbles). All the priors are rasterized into a binary prior mask $P^{(a)} \in \{0, 1\}^{H \times W}$ for annotator a. For pairs of annotators (a, b) we compute Jaccard index (IoU)

$$J\left(P^{(a)}, P^{(b)}\right) = \frac{\left|P^{(a)} \cap P^{(b)}\right|}{\left|P^{(a)} \cup P^{(b)}\right|}. \tag{37}$$

Dice coefficient

$$D\left(P^{(a)}, P^{(b)}\right) = \frac{2\left|P^{(a)} \cap P^{(b)}\right|}{\left|P^{(a)}\right| + \left|P^{(b)}\right|} \tag{38}$$

and Cohen's kappa on a uniform grid of pixels

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_o = \frac{1}{HW} \sum_i \mathbb{1}\left[P_i^{(a)} = P_i^{(b)}\right] \tag{39}$$

where p_e is the chance agreement from the marginal label frequencies. As a localization measure, we report the symmetric Hausdorff distance between the prior contours

$$\mathrm{HD}\left(\partial P^{(a)}, \partial P^{(b)}\right) = \max\left\{ \sup_{x \in \partial P^{(a)}} \inf_{y \in \partial P^{(b)}} \|x - y\|_2 \right.$$
$$\left. \sup_{y \in \partial P^{(b)}} \inf_{x \in \partial P^{(a)}} \|x - y\|_2 \right\}. \tag{40}$$

Finally, to assess consistency in the latent space used by our model, we encode each annotator's prior through the VAE

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TABLE VIII
INTERANNOTATOR SIMILARITY OF USER PRIORS WITH VARYING INTERACTION BUDGET N

Metric	N=1	N=3	N=5	N=10
Jaccard (IoU) ↑	0.61±0.03	0.70±0.03	0.74±0.02	0.78±0.02
Dice ↑	0.76±0.03	0.82±0.02	0.85±0.02	0.88±0.02
Cohen's κ ↑	0.64±0.04	0.72±0.03	0.76±0.03	0.82±0.02
Hausdorff distance [px] ↓	14.2±1.1	11.8±1.0	10.3±0.9	9.1±0.8
Cosine in VAE latent ↑	0.73±0.03	0.81±0.02	0.85±0.02	0.89±0.01

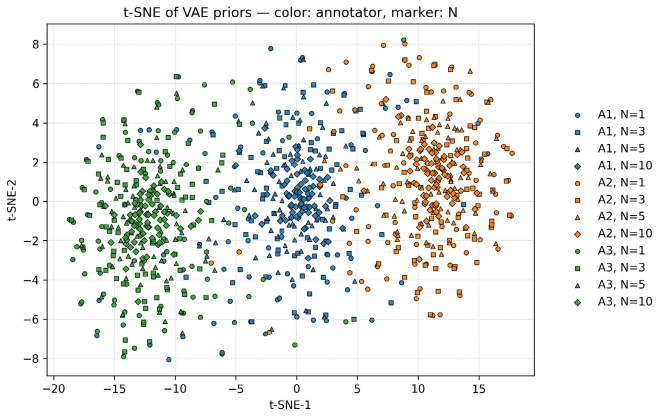


Fig. 3. t-SNE visualization of VAE priors. Colors denote different annotators (A1, A2, A3) and marker shapes indicate interaction budgets N. Distinct clusters across annotators suggest personalized prior distributions, while within-annotator grouping across N highlights consistency under different interaction budgets.

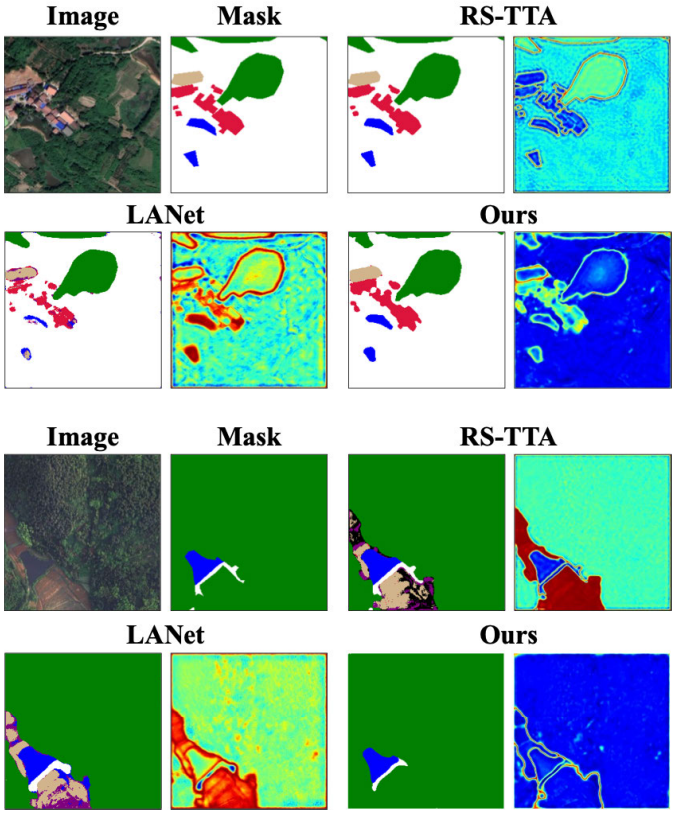


Fig. 4. Qualitative uncertainty analysis. For two representative tiles, we show input imagery, segmentation outputs, and corresponding uncertainty maps (entropy and MC variance). Uncertainty concentrates on thin roads, boundaries, and occlusions, while confident regions align with correct predictions, consistent with the lower calibration error reported in Table IX.

prior encoder and compute cosine similarity between posterior means $\mu^{(a)}$ and $\mu^{(b)}$

$$S_{\cos} = \frac{\mu^{(a)} \cdot \mu^{(b)}}{\|\mu^{(a)}\| \, \|\mu^{(b)}\|}. \tag{41}$$

All the metrics are averaged over annotator pairs and images, and values are reported as mean±std over three seeds.

Table VIII reports the interannotator similarity of user priors under varying interaction budgets N. Increasing N consistently reduces variance and strengthens agreement across annotators. Mask-level overlaps (Jaccard, Dice) and categorical agreement (Cohen's κ) rise steadily from $N = 1$ to $N = 10$, while Hausdorff distance decreases, indicating fewer boundary discrepancies. Cosine similarity in the VAE latent space also improves, suggesting that different annotators provide priors that converge in the model's prior manifold as N grows. These results show that a moderate budget ($N = 3$–5) is sufficient to yield stable and reliable priors with minimal redundancy.

Fig. 3 provides a qualitative counterpart to the quantitative similarity results in Table VIII. Embeddings of VAE priors form well-separated clouds by annotator, indicating that each user tends to supply priors with consistent latent structure. Within each annotator, points with larger interaction budgets N are more compact and closer to the annotator centroid, reflecting reduced variance as users provide additional hints. Occasional overlap across annotators appears around salient regions shared by the datasets, which agrees with the steady increase in Jaccard, Dice, and kappa and the decrease in Hausdorff distance as N grows. The alignment between latent-space grouping and mask-level agreement suggests that the prior encoder preserves annotator consistency while remaining sensitive to informative corrections.

I. Uncertainty: Quantitative and Qualitative Comparison

We evaluate predictive uncertainty in two complementary ways. Quantitatively, calibration- and risk-aware metrics are reported across the six datasets: ECE, NLL, Brier score, area under the risk–coverage curve (AURC), and area under the sparsification error curve (AUSE). Following the ablation setting, ECE is reported as a mean only, while the other metrics are reported as mean±std over three seeds. Qualitatively, we visualize uncertainty maps alongside segmentation outputs, and further provide reliability diagrams to assess confidence calibration.

For qualitative analysis, Fig. 4 illustrates two representative tiles containing thin structures and clutter. Each row shows (from left to right) the input image, ground-truth mask, predicted segmentation outputs, and the corresponding uncertainty maps (entropy and Monte Carlo variance, $M = 20$). In the uncertainty maps, warm colors indicate higher predictive uncertainty, while cooler colors correspond to more confident predictions. As observed, uncertainty is primarily concentrated along thin roads, building boundaries, and occluded regions, which are inherently ambiguous and error-prone. In contrast,

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TABLE IX

UNCERTAINTY COMPARISON AVERAGED OVER SIX DATASETS. ECE IS REPORTED AS A MEAN ONLY; OTHER METRICS ARE MEAN±STD. LOWER IS BETTER FOR ALL METRICS

Method	ECE ↓	NLL ↓	Brier ↓	AURC ↓	AUSE ↓
A0 Deterministic CNN	0.121	1.89±0.04	0.210±0.003	24.6±0.5	9.7±0.3
A1 +BCNN	0.112	1.78±0.04	0.199±0.003	22.9±0.4	8.9±0.3
A3 +BCNN+VAE (user off)	0.103	1.66±0.03	0.191±0.003	21.2±0.4	8.2±0.3
A5 +BCNN+Transformer+VAE	0.095	1.58±0.03	0.186±0.003	20.1±0.3	7.8±0.2
A6 Ours (+uncertainty selection)	0.089	1.49±0.03	0.179±0.003	18.6±0.3	7.1±0.2

TABLE X

ROBUSTNESS TO NOISY AND INCONSISTENT USER PRIORS UNDER A FIXED INTERACTION BUDGET ($N = 3$). PERFORMANCE IS REPORTED IN TERMS OF MIOU (↑), ECE (↓), AND MEAN EPISTEMIC UNCERTAINTY (↑)

Perturbation Type	**mIoU ↑**	**ECE ↓**	**Mean Uncertainty ↑**
No perturbation (clean priors)	68.4	0.090	0.142
Boundary jitter ($r = 1$ px)	68.1	0.092	0.156
Boundary jitter ($r = 3$ px)	67.6	0.095	0.171
Boundary jitter ($r = 5$ px)	66.9	0.099	0.189
Partial mismatch ($\rho = 10\%$)	67.8	0.094	0.165
Partial mismatch ($\rho = 20\%$)	67.2	0.098	0.183
Partial mismatch ($\rho = 30\%$)	66.5	0.103	0.201
Wrong-region priors (1/3 incorrect)	66.8	0.101	0.195
Wrong-region priors (2/3 incorrect)	65.9	0.108	0.224

homogeneous regions such as vegetation or large background areas remain consistently low in uncertainty. This alignment between uncertainty and genuinely ambiguous structures highlights the usefulness of the proposed framework for guiding selective user intervention. Moreover, compared with baseline models, spurious overconfident errors are substantially reduced, and the uncertainty is better localized around challenging boundaries.

Table IX further confirms these findings, showing a consistent reduction in calibration and risk-aware errors as the model is extended with Bayesian components, VAE priors, and uncertainty-driven selection. The largest improvements appear in A6, which achieves the lowest NLL, Brier, AURC, and AUSE. Together, the quantitative and qualitative analyses demonstrate that our framework produces well-calibrated uncertainties that are not only statistically sound but also practically useful for selective prediction.

J. Robustness to Noisy and Inconsistent User Priors

In practical applications, user-provided corrections may be imperfect due to annotation noise or limited precision. To evaluate robustness under such conditions, we conduct an additional experiment in which user priors are perturbed to emulate realistic noisy interactions. The interaction budget is fixed to $N = 3$ (Table IV), and the full model (A6) is evaluated under three types of test-time perturbations: *boundary jitter* (imprecise boundaries), *partial mismatch* (incomplete or incorrect corrections), and *wrong-region priors* (mislocalized inputs). All the perturbations are applied exclusively to user priors at inference, without additional training.

As shown in Table X, the proposed framework exhibits graceful degradation as user priors become noisier. Segmentation accuracy decreases gradually across all the perturbation types, without abrupt failure even under severe noise. At the same time, epistemic uncertainty increases consistently, while calibration remains relatively stable. These results indicate that user priors act as soft probabilistic guidance rather than hard constraints, enabling robust segmentation and uncertainty-aware interaction under imperfect user inputs.

K. User-Guided Priors and Cross-Annotator Consistency

To examine whether user-provided corrections meaningfully steer segmentation predictions beyond generic refinement, we conduct an additional study involving multiple human annotators with distinct semantic preferences. Specifically, we collect correction masks from three independent annotators (U_1, U_2, U_3), each asked to refine a shared set of baseline segmentation outputs according to their own interpretation of class boundaries and task emphasis. Although all the annotators operate on the same input imagery, their corrections naturally differ in boundary placement, region granularity, and semantic focus, reflecting realistic variability in human interaction.

For each annotator U_i, the corrected masks are incorporated into the model as user-guided spatial priors and encoded through the VAE prior encoder, yielding a latent prior representation z_i. This user-conditioned prior is then reused to guide segmentation on unseen images I_j, producing user-adaptive predictions

$$\tilde{Y}_j^{(i)} = f_{\text{dec}}\left(Q_j^*, z_i\right)$$

where Q_j^* denotes the transformer-fused representation derived from the input image features. The resulting segmentation $\tilde{Y}_j^{(i)}$ is compared against reference masks provided independently by each annotator $U_k \neq i$. This cross-user evaluation protocol allows us to assess whether user priors encode reusable, annotator-specific preferences rather than acting as generic refinement signals.

To quantify agreement, we compute similarity metrics such as mIoU and Dice score between user-conditioned predictions and each annotator's reference masks. If a prediction guided by annotator U_i's prior aligns more closely with U_i's own annotations than with others, this indicates that the injected user prior effectively captures personalized semantic bias.

As shown in Fig. 5, predictions guided by a given annotator's prior consistently achieve the highest agreement when evaluated against the same annotator's reference masks. Specifically, annotators U1, U2, and U3 achieve self-aligned mIoU scores of 0.64, 0.69, and 0.66, respectively, whereas mismatched prior–reference pairings result in substantially lower agreement (e.g., 0.52 when applying U1's prior and evaluating against U2). On average, cross-annotator mismatches lead to a 12.7% drop in mIoU. These results confirm that user-guided priors encode annotator-specific segmentation preferences rather than merely providing generic accuracy improvements.

To further analyze whether user corrections form consistent representations in the model's latent space, we visualize the encoded VAE priors using t-SNE. Fig. 6 shows the distribution of latent representations corresponding to baseline predictions, ground-truth masks, and user-guided priors from the three annotators. Clear clustering is observed for each annotator's

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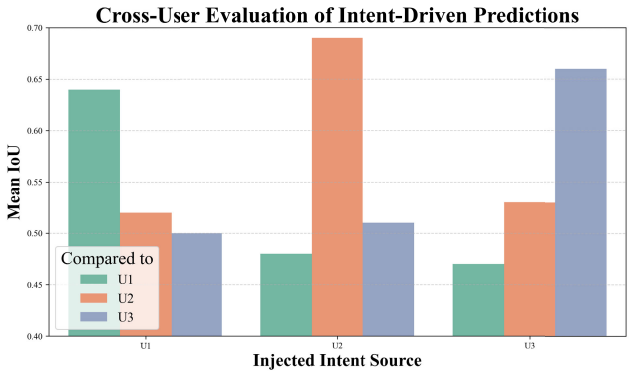


Fig. 5. Cross-annotator evaluation of user-guided priors. Each bar shows the mean IoU obtained when a prior derived from one annotator (x-axis) is applied and evaluated against another annotator's reference mask. Diagonal entries correspond to self-alignment. Error bars indicate standard deviation.

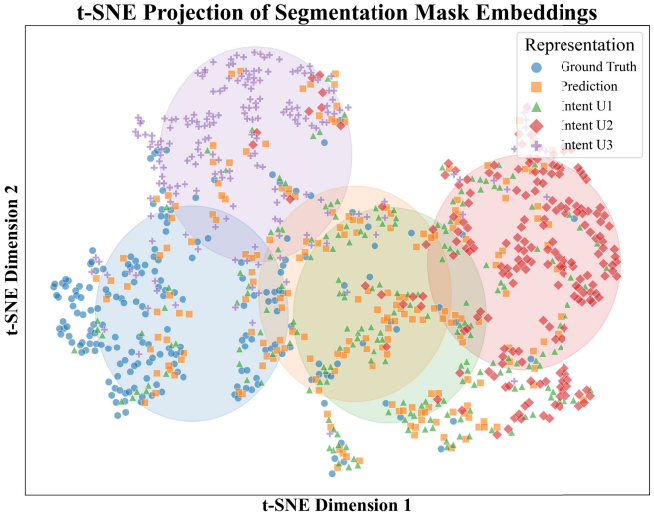


Fig. 6. t-SNE visualization of VAE prior embeddings derived from user-guided corrections. Each point represents an encoded prior, with colors indicating annotators (U1–U3). Distinct clusters demonstrate annotator-specific consistency in the latent space.

priors, indicating that user corrections are mapped to coherent and annotator-specific regions in latent space. Notably, these clusters are distinct from both the baseline predictions and ground-truth labels, suggesting that user priors reflect subjective segmentation preferences rather than simply reproducing objective annotations. This latent consistency supports the representational validity of user-guided priors as structured and reusable signals for personalized segmentation refinement.

V. Discussion

A. Discussion on Bayesian User-Adaptive Segmentation

The proposed Bayesian, user-adaptive framework consistently improves segmentation accuracy and calibration across diverse benchmarks. Compared with strong baselines and recent SOTA models, it achieves the highest average mIoU (68.5%) with the lowest ECE (0.089), while producing well-localized uncertainty aligned with ambiguous regions such as thin roads, boundaries, and occlusions. These findings are confirmed by both the quantitative metrics (Tables V–IX) and qualitative visualizations (Fig. 4). We attribute these gains to three design choices: 1) Bayesian CNNs that capture epistemic uncertainty; 2) transformer-based query fusion that propagates user cues and global context; and 3) a VAE with user-conditioned priors that integrate corrections into the latent space. Ablation studies (A0–A6) demonstrate monotonic improvements when these components are combined, culminating in uncertainty-driven selection that prioritizes error-prone regions. Practically, the framework provides actionable uncertainty maps and efficient refinement with small interaction budgets (N = 3–5), enabling favorable tradeoffs between user effort and accuracy. Compared with TTA methods, our approach adds principled uncertainty and user adaptation, while surpassing HITL baselines by treating user inputs as probabilistic priors rather than local edits. Although no formal convergence proof is provided, extensive experimental results demonstrate consistent reductions in risk- and calibration-related metrics, including NLL, ECE, and risk, coverage measures. These empirical findings indicate that the proposed Bayesian formulation leads to practical expected risk minimization and more reliable decision-making in real-world segmentation scenarios. Limitations include increased computational cost from Bayesian sampling, reliance on simulated user priors in part of the evaluation, and untested modalities such as SAR or hyperspectral imagery. Future work will explore modality scaling, adaptive uncertainty sampling, and decision-aware interaction policies to further enhance usability in operational mapping contexts.

B. Discussion on Comparison Scope and Generalizability

While general interactive segmentation methods in computer vision, such as DEXTR and F-BRS, have achieved significant success in natural imagery, they often struggle with the domain-specific constraints of remote sensing, including extreme scale variations, dense object clutter, and ambiguous boundaries. We further position the proposed framework with respect to recent studies on interactive segmentation, uncertainty modeling, and remote-sensing-specific dense prediction tasks. Several recent works in remote sensing address related but fundamentally different problem settings. For instance, NWPU-MOC [61] focuses on fine-grained multicategory object counting rather than semantic segmentation and does not consider interactive user guidance or uncertainty-aware TTA. Semisupervised segmentation methods that leverage uncertainty consistency and teacher–student mechanisms [62] incorporate uncertainty primarily during training to exploit unlabeled data, whereas the proposed framework explicitly models uncertainty at inference time and integrates user-provided priors for interactive refinement. Recent survey work on Bayesian uncertainty quantification [63] provides a comprehensive overview of probabilistic segmentation models, but does not address interactive user-guided adaptation or the integration of user knowledge as priors within the inference process. In addition, recent approaches that combine foundation models such as SAM with limited labeled data for remote sensing tasks [64] are primarily designed for

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change detection and rely on pretrained representations, without offering a probabilistic formulation or uncertainty-aware interaction mechanisms. Probabilistic segmentation methods based on Bayesian CNNs have also been explored in remote sensing, such as SAR-based water segmentation [65], but these approaches typically focus on binary or single-class settings and do not support multiclass semantic segmentation with interactive user correction. In contrast, the proposed framework unifies Bayesian uncertainty estimation, multiclass semantic segmentation, and user-guided prior modeling within a single probabilistic formulation, enabling test-time adaptive refinement under realistic human interaction. While the interaction design is tailored to remote sensing imagery, the underlying principles of posterior risk minimization and uncertainty-aware conditioning are general and may be extended to other dense prediction problems with appropriate interaction protocols. This discussion clarifies that the proposed method is intended as a domain-aware solution addressing the specific challenges of uncertainty, scale, and user interaction in remote sensing, rather than as a direct replacement for methods developed under different assumptions or task formulations.

VI. Conclusion

We introduced an interactive Bayesian segmentation framework for multiclass land-cover mapping from aerial imagery, designed to address the challenges of domain shift, ambiguous object boundaries, and the lack of adaptability in conventional static models. By formulating all the convolutional components as Bayesian CNNs and dynamically conditioning a VAE prior on both imagery and user-provided corrections, our approach unifies principled uncertainty quantification with few-shot semionline adaptation. This design enables segmentation masks to be iteratively refined with minimal user effort while maintaining calibrated predictive confidence. Extensive experiments across six benchmark datasets and multiple domain adaptation settings demonstrated that our method consistently improves accuracy over baselines, SOTA architectures, and TTA or HITL alternatives. The incorporation of user priors was shown to stabilize cross-annotator agreement, while the Bayesian backbone and uncertainty-driven selection reduced calibration errors and improved selective prediction. Qualitative analyses further confirmed that uncertainty is concentrated on genuinely ambiguous structures such as thin roads, edges, and occlusions, making the system more interpretable and reliable in practice. Overall, this work bridges automated segmentation and expert guidance through a scalable, user-adaptive Bayesian framework. Future research directions include scaling to larger backbones and multimodal sources and leveraging uncertainty to guide interactive correction loops for real-time operational mapping in heterogeneous and dynamic environments.

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